



Retail Sales & Profitability

End-to-end analysis (2014–2017): Python → SQL Server → Power BI



Dataset at a glance

Rows

9,994 order line items

Columns

21 raw → 24 after feature engineering

Date Range

Jan 3, 2014 – Dec 30, 2017

Data Quality

0 missing, 0 duplicates

Key features



Customer

customer_id, name, segment



Product

product_id, category, sub_category



Geography

city, state, postal_code, region



Metrics

sales, quantity, discount, profit



Python — preparation & checks

01

Load & inspect

`read_excel`, `df.info()`, `describe`, `isnull()`

02

Standardize

rename to `snake_case`

03

Integrity

zero nulls, zero duplicates, `ship_date` $\geq$ `order_date`

04

Features

`profit_margin`, `shipping_days`, `order_year_month`, `is_loss`

Discounts drive losses



Below 40% discount

Order count: 8,855 — Profit margin: 19.93%

40% or more

Order count: 1,139 — Profit margin: -50.04%

Outlier check: bottom 1% all $\geq 40\%$ discount

Star schema (SQL Server)

1

Fact_Orders

order_id, sales, quantity, discount, profit

2

Dim_Customer

customer_id, customer_name, segment

3

Dim_Product

product_id, product_name, category, sub_category

4

Dim_Geography

geo_id, postal_code, city, state, region

5

Dim_Date

date, year, month, quarter, year_month

Key SQL analyses



Monthly sales & profit

Seasonal trends
across 2014–2017



Top sub-categories

Ranked by profit
margin



Loss-making sub-categories

Tables, Bookcases,
Supplies (neg. total
profit)



Top customers

Sean Miller
\$25,043; Tamara
Chand \$19,052;
Raymond Buch
\$15,117

Operations & shipping



Shipping performance

Same Day ~0.04 days; Standard Class ~5.01 days

Impact

Standard Class = largest volume, opportunity to improve CX



Executive dashboard (Power BI)



KPIs

Total Sales \$2.30M • Total Profit \$286K • Margin 12.47%

Pages

Executive, Sales & Profitability, Customers, Operations

Model: Fact_Orders + 4 dims • DAX measures for time intelligence

Insights & recommendations



Limit deep discounts

Cap discounts >30–40% or require approval (40%+ is unprofitable)



Fix Tables

Review pricing/sourcing for Tables (largest loss driver)



Invest West & East

Expand inventory/marketing where profit is strongest



Prioritize Consumer

Retention campaigns for highest-profit segment



Improve Standard Class

Reduce avg. 5-day fulfillment to lift CX